

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 9

Aim: Conducting Chi-square tests using `chisq.test()` (R)

Code:

```
library(tidyverse)

data <- read.csv("meteorite-landings.csv")

data <- data %>% filter(!is.na(mass), !is.na(year))

median_revenue <- median(data$mass)
median_net_income <- median(data$year)

data$Revenue_Cat <- ifelse(data$mass >= median_revenue, "High", "Low")
data$Net_Income_Cat <- ifelse(data$year >= median_net_income, "High", "Low")

observed <- table(data$Revenue_Cat, data$Net_Income_Cat)
print("Contingency Table (Observed):")
print(observed)

row_totals <- rowSums(observed)
col_totals <- colSums(observed)
grand_total <- sum(observed)

expected <- outer(row_totals, col_totals) / grand_total

chi_sq_stat <- sum((observed - expected)^2 / expected)

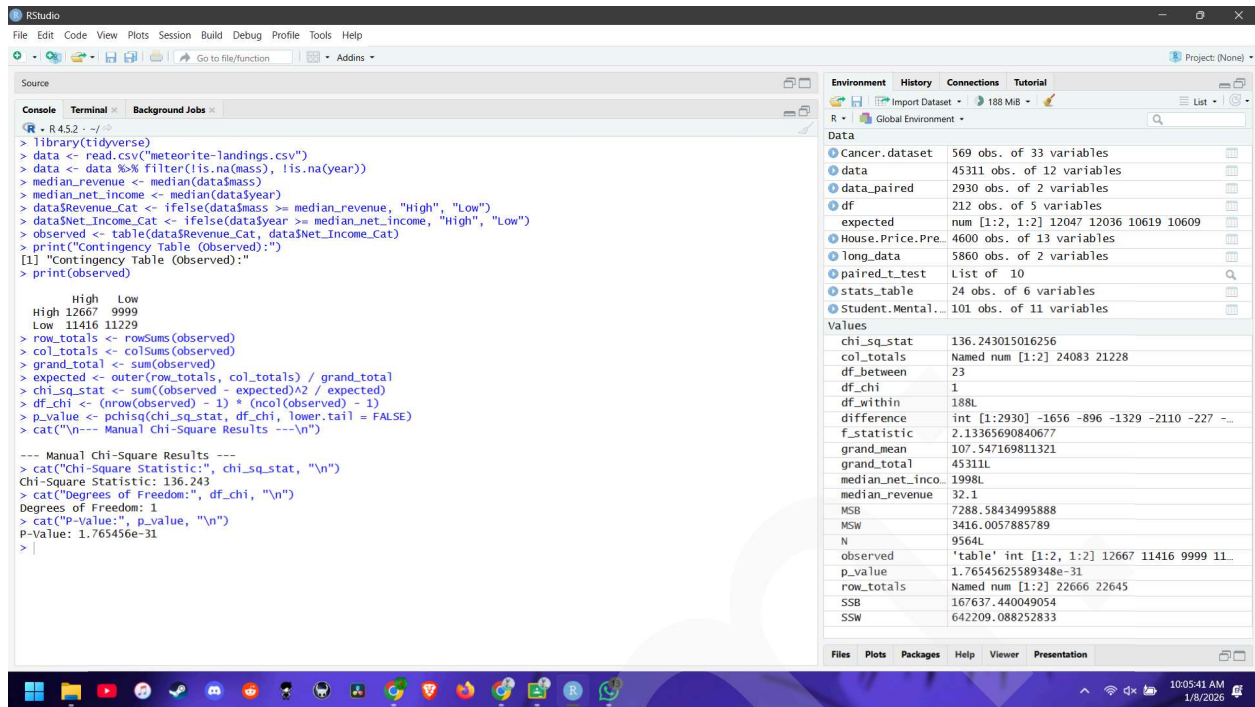
df_chi <- (nrow(observed) - 1) * (ncol(observed) - 1)

p_value <- pchisq(chi_sq_stat, df_chi, lower.tail = FALSE)

cat("\n--- Manual Chi-Square Results ---\n")
cat("Chi-Square Statistic:", chi_sq_stat, "\n")
cat("Degrees of Freedom:", df_chi, "\n")
cat("P-Value:", p_value, "\n")
```

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Output:



The screenshot shows the RStudio interface with the following content:

```
R > R4.5.2 - ~/ -  
File Edit Code View Plots Session Build Debug Profile Tools Help  
Go to file/function Addins  
Source  
Console Terminal Background Jobs  
R > library(tidyverse)  
> data <- read.csv("meteorite-landings.csv")  
> data <- data %>% filter(!is.na(mass), !is.na(year))  
> median_revenue <- median(data$mass)  
> median_net_income <- median(data$year)  
> data$Revenue_Cat <- ifelse(data$mass >= median_revenue, "High", "Low")  
> data$Net_Income_Cat <- ifelse(data$year >= median_net_income, "High", "Low")  
> observed <- table(data$Revenue_Cat, data$Net_Income_Cat)  
> print("Contingency Table (observed):")  
[1] "Contingency Table (observed):"  
> print(observed)  
      High Low  
High 12667 9999  
Low 11416 11229  
> row_totals <- rowSums(observed)  
> col_totals <- colSums(observed)  
> grand_total <- sum(observed)  
> expected <- outer(row_totals, col_totals) / grand_total  
> chi_sq_stat <- sum((observed - expected)^2 / expected)  
> df_chi <- (nrow(observed) - 1) * (ncol(observed) - 1)  
> p_value <- pchisq(chi_sq_stat, df_chi, lower.tail = FALSE)  
> cat("\n--- Manual Chi-Square Results ---\n")  
--- Manual Chi-Square Results ---  
> cat("Chi-Square Statistic:", chi_sq_stat, "\n")  
Chi-Square Statistic: 136.243  
> cat("Degrees of Freedom:", df_chi, "\n")  
Degrees of Freedom: 1  
> cat("P-Value:", p_value, "\n")  
P-Value: 1.765456e-31  
>
```

The Environment pane on the right shows the following data objects:

Object	Size
Cancer.dataset	569 obs. of 33 variables
data	45311 obs. of 12 variables
data_paired	2930 obs. of 2 variables
df	212 obs. of 5 variables
expected	num [1:2, 1:2] 12047 12036 10619 10609
House.Price.Pre	4600 obs. of 13 variables
long_data	5860 obs. of 2 variables
paired_t_test	List of 10
stats_table	24 obs. of 6 variables
Student.Mental...	101 obs. of 11 variables

The Values pane shows the following results:

Variable	Value
chi_sq_stat	136.243015016256
col_totals	Named num [1:2] 24083 21228
df_between	23
df_chi	1
df_within	188L
difference	int [1:2930] -1656 -896 -1329 -2110 -227 ...
f_statistic	2.13365690840677
grand_mean	107.547169811321
grand_total	45311L
median_net_inco	1998L
median_revenue	32.1
MSB	7288.58434995888
MSW	3416.0057885789
N	9564L
observed	'table' int [1:2, 1:2] 12667 11416 9999 11...
p_value	1.76545625589348e-31
row_totals	Named num [1:2] 22666 22645
SSB	167637.440049054
SSW	642209.088252833